

Alan Arimpur Raju | Data Scientist | *Birmingham, UK*

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Education

MSc Data Science (Distinction) - University of Birmingham, Birmingham, UK | *Graduating Dec 2025*

Applied Skills: Predictive Modelling | NLP | Time Series Forecasting | Risk Analytics | Python | SQL | PyTorch | Financial Data Analysis | Machine Learning | Deep Learning | Statistical Analysis

BTech Electronics & Communication - Amrita Vishwa Vidyapeetham, Bangalore, India | *Jun 2025*

Applied Skills: Python | MATLAB | SQL | Signal Processing | Embedded Systems | Data Visualization

Experience

BEAR Challenge - *University of Birmingham, Birmingham, UK* | Jun 2025

- Implemented SLURM job scripts on the Baskerville supercomputer to deploy a CUDA optimized password-hash cracking program across eight GPUs on two nodes, enabling near-instant password cracking.
- Refined hyperparameters on the Baskerville supercomputer for the Zoobot deep learning galaxy image classifier, achieving 90% test accuracy.

Consultancy Challenge - *Turner & Townsend (UoB), Birmingham, UK* | Jun 2025

- Directed a 6-member interdisciplinary team as a consultant, applying MBA project management tools to set goals and deliver work efficiently and on time.
- Developed and presented data-driven digital-first sustainability strategies to University of Birmingham and Turner & Townsend stakeholders, analyzed travel and energy data to identify high-impact emission drivers, and proposed solutions with potential £2M cost savings.

Associate - *Cognizant, Bangalore, India* | Nov 2021 - June 2024

- Automated backup validation and reporting workflows for multi-OS server environments, enhancing operational efficiency and reducing incidents by 40% using Python and Bash, with a focus on data integrity.
- Collaborated with cross-functional teams and external vendors to replace faulty DataDomain servers, ensuring uninterrupted data operations in remote data centers.
- Built a high-quality MERN Stack project with seamless API integration, demonstrating full-stack data handling and earning recognition as a benchmark for peer projects.

Projects

Portfolio Risk Analysis Dashboard | *Github: <https://github.com/ReptilianPride/portfolio-risk-dashboard>*

Python, Pandas, Numpy, Plotly, Streamlit, Risk Analysis, Monte Carlo

- Developed a Python-based portfolio risk dashboard with VaR, Sharpe Ratio, Drawdowns, and Volatility, integrated Monte Carlo simulations to optimize portfolio weights, and benchmarked performance against the S&P 500 for data-driven decisions.

Forecasting Emerging News Trends | *Masters Dissertation*

Python, NLP, Network Science, Statistical Modelling, Sentence Transformers, FAISS, cuGraph

- Engineered a scalable Python pipeline to process 190k global news articles, leveraging NLP with sentence transformers and network science to extract patterns and key features for trend prediction.
- Delivered high-impact predictive models with 81% accuracy on entity-based classification and 8.3% error in time-series forecasting, enabling early, interpretable identification of emerging news narratives.

Data Driven study on Tweets Influencing Stock Market | *Group Project*

Python, Pandas, Numpy, Machine Learning, Web Scraping

- Contributed to an open-source project to enable more stable collection of stock-related tweets for analysis.
- Analyzed 100,000+ stock-related tweets using sentiment analysis, PCA, and classical ML models, leading a team to find that tweet sentiment has a weak and inconsistent influence on stock prices.

Awards & Extracurricular

- Received The Birmingham Award by the University of Birmingham.
- Secretary of FORGE Club (Graphics Design Club).
- Trinity college London certified for Drums.